

Assignment No 02

Name: Om Varma

Roll No: 28

PRN No: 1252010028

Question:

Constructor and destructor (Book)

Code:

```
#include<iostream>
using namespace std;

class Books{
public:
    long* bookID;
    long* price;
    string* authorName;

    long validateID(long id){
        if(id <= 0){
            return 1;
        }
        return id;
    }

    long validatePrice(long p){
        if(p <= 0){
            return 100;
        }
        return p;
    }

    string validateAuthor(string name){
        if(name.length() == 0){
            return "Unknown";
        }
        return name;
    }

    Books(){
        bookID = new long;
        price = new long;
        authorName = new string;

        cout << "Constr invoked" << endl << endl;

        *bookID = 1;
        *price = 1000;
        *authorName = "Om Varma";
    }
}
```

```

    ~Books(){
        delete bookID;
        delete price;
        delete authorName;
        cout << "Destr invoked" << endl << endl;
    }

    void input(){
        long id;
        long p;
        string name;

        cout << "Enter Book ID: ";
        cin >> id;

        cout << "Enter Price: ";
        cin >> p;

        cin.ignore();
        cout << "Enter Author Name: ";
        getline(cin, name);

        *bookID = validateID(id);
        *price = validatePrice(p);
        *authorName = validateAuthor(name);
    }

    void display(){
        cout << "Book ID:" << *bookID << endl;
        cout << "Price:" << *price << endl;
        cout << "Author Name:" << *authorName << endl << endl;
    }
};

int main(){
    int n;

    cout << "Enter number of books: ";
    cin >> n;

    Books* arr = new Books[n];

    for(int i = 0; i < n; i++){
        cout << "Enter details for book " << i + 1 << endl;
        arr[i].input();
    }

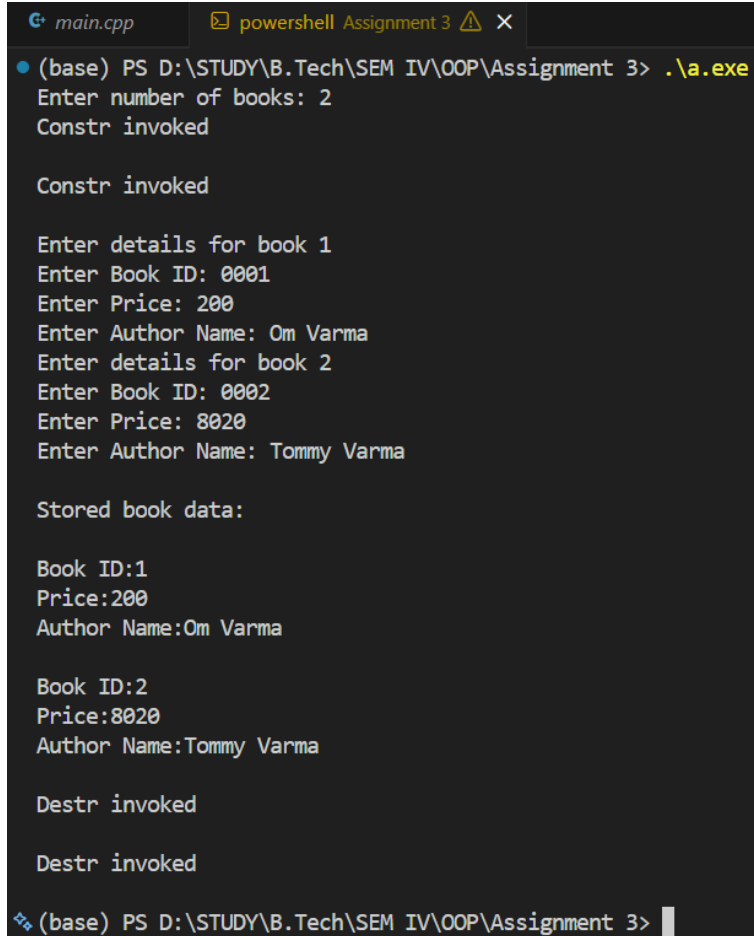
    cout << endl << endl << "Books:" << endl << endl;

    for(int i = 0; i < n; i++){
        arr[i].display();
    }
}

```

```
delete[] arr;  
}
```

Output:



```
(base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 3> .\a.exe  
Enter number of books: 2  
Constr invoked  
  
Constr invoked  
  
Enter details for book 1  
Enter Book ID: 0001  
Enter Price: 200  
Enter Author Name: Om Varma  
Enter details for book 2  
Enter Book ID: 0002  
Enter Price: 8020  
Enter Author Name: Tommy Varma  
  
Stored book data:  
  
Book ID:1  
Price:200  
Author Name:Om Varma  
  
Book ID:2  
Price:8020  
Author Name:Tommy Varma  
  
Destr invoked  
  
Destr invoked  
  
❖ (base) PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 3> |
```